

PROJECT REPORT

BSDA4001: DATA SCIENCE AND AI LAB

APRIL 2026



Multimodal AI Assistant for Smart Agriculture

Crop Disease Detection | Grounded Advisory | Multilingual Voice Support

Team 8

Harish Sahadev M	21f1005856
Sai Naman	22f3000350
Ayushi Dixit	22f3000082
Manas Rastogi	22f3001477
Saketh Allanki	21f3002277

Table of Contents

1. Abstract	3
2. Introduction	4
2.1 Background	4
2.2 Problem Statement	4
2.3 Objectives	4
2.4 Scope of the Project	4
3. Literature Review	5
3.1 Deep Learning for Crop Disease Detection	5
3.2 Retrieval-Augmented Generation in Domain-Specific AI	5
3.3 Multilingual and Voice-Based AI Systems	5
3.4 Existing Solutions and Research Gaps	5
4. Dataset and Methodology	6
4.1 Dataset Overview	6
4.2 Crop and Class Composition	6
4.3 EDA	7
4.4 Preprocessing Pipeline	7
4.5 Augmentation Strategy... ..	7
4.6 Data Split and Pipeline Verification	8
4.7 Initial Architecture Exploration	9
5. Model Development and Hyperparameter Tuning	10
5.1 Model Selection	10
5.2 Transfer Learning Strategy	10
5.3 Hyperparameter Search using Optuna	10
5.4 Best Model Configuration	10
5.5 Training Results	11
6. Evaluation and Analysis	12
6.1 Evaluation Step	12
6.2 Quantitative performance	12
6.3 Class-wise Analysis	12
6.4 Confusion Matrix and Error Analysis	13
6.5 Qualitative Results	13
6.7 Key Evaluation Insights	15
7. Deployment and Documentation	15
7.1 System Overview	15
7.2 End-to-end pipeline	15
7.3 RAG Stack	15
7.4 Voice and Multilingual Features	16
7.5 Deployed Constraints and Design Trade-Offs	16
7.6 Documentation Delivered	16
7.7 User-Facing Safety Measures	16
8. Conclusion and Future Work	17
8.1 Conclusion	17

8.2 Limitations	17
8.3 Future Work	17
8.4 Final Remark	17
Team Signatures	18

1. ABSTRACT

Agriculture remains highly vulnerable to disease outbreaks that can sharply reduce yield, increase input cost, and threaten the income of small and marginal farmers. In practice, timely diagnosis is difficult because expert support is often delayed, digital tools are frequently English-centric, and generic AI systems may produce unsafe or hallucinated treatment advice. This project addresses those gaps through a unified multimodal system that combines computer vision, retrieval-augmented generation (RAG), multilingual interaction, and voice support in a single deployable application.

The system classifies crop leaf images into 17 health and disease categories across five crops: corn, potato, rice, wheat, and sugarcane. The vision backbone is a transfer-learned MobileNet V3 Large model selected through Optuna-based hyperparameter optimization with Macro F1 as the primary objective to account for severe class imbalance. On the held-out test set, the final classifier achieved 92.09% accuracy and 90.04% Macro F1, while maintaining a lightweight footprint suitable for low-cost deployment.

To make recommendations trustworthy, the system uses a RAG pipeline built over verified agricultural documents from ICAR, TNAU, and FAO. Instead of allowing a large language model to answer from memory, the application retrieves relevant crop-disease passages and constrains the generated advisory to the retrieved evidence. A multilingual voice interface based on Whisper ASR and gTTS enables interaction in English, Hindi, Bengali, Tamil, Telugu, Malayalam, and Kannada. The full application is deployed on Hugging Face Spaces using free-tier infrastructure, demonstrating that a practical, farmer-facing AI assistant can be built with strong technical performance, grounded advisory behavior, and accessible user experience.

2. INTRODUCTION

2.1 Background

Crop diseases such as blights, rusts, and bacterial infections can spread rapidly, especially when early symptoms are missed. For many farmers, the challenge is not only identifying the disease correctly but doing so quickly enough to take action before the infection affects a large part of the field. In India, this challenge is amplified by unequal access to agricultural experts, language diversity, and inconsistent network connectivity. Existing digital solutions often solve only one part of the problem: some classify images, some provide advisory chat, and some support local languages, but very few combine all of them in one system.

2.2 Problem Statement

There is a clear need for an end-to-end agricultural assistant that can:

- Detect crop disease directly from a leaf image
- Provide reliable treatment and prevention advice
- Support multilingual and voice-first interaction
- Avoid hallucinated recommendations, and
- Run on low-cost infrastructure suitable for practical demonstration and outreach.

2.3 Project Objectives

Primary objective:

Design and deploy a multimodal AI system that detects crop diseases from leaf images and provides grounded, multilingual, voice-enabled advisory support.

Secondary objectives:

- Achieve strong classification performance on a 17-class crop disease dataset
- Address class imbalance through preprocessing and augmentation
- Ground advisory responses in verified agricultural documents using RAG
- Support seven response languages including English and six Indian languages
- Integrate speech input and output for accessibility
- Deploy the complete pipeline on free-tier, CPU-constrained infrastructure

2.4 Overall System Contribution

The project contributes a complete multimodal pipeline rather than an isolated classifier. It combines:

1. Deep learning vision model for disease detection
2. Metadata-aware RAG subsystem for advisory retrieval
3. LLM for response generation under grounding constraints
4. Multilingual speech interface for natural interaction, and
5. Web deployment that is immediately usable by end users.

3. LITERATURE REVIEW

3.1 Deep Learning for Crop Disease Detection

Plant disease classification has shifted from handcrafted color and texture features to transfer learning with convolutional neural networks. Benchmark datasets such as PlantVillage established strong baseline performance for architectures including ResNet, MobileNet, and EfficientNet under controlled conditions. However, reported accuracy on curated datasets does not always translate to field settings, where lighting, background clutter, occlusion, and viewpoint variation are common. This gap motivated our emphasis on preprocessing discipline, class-aware augmentation, and deployment-aware model selection.

3.2 Severity Estimation and Fine-Grained Recognition

Prior work on severity estimation often relies on segmentation or specialized severity labels, neither of which is consistently available in public agricultural datasets. While severity prediction was discussed during early project planning, the final system prioritized reliable disease-category recognition and grounded advisory generation because the chosen dataset did not provide robust, uniformly labeled severity annotations across all crops. This decision kept the project aligned with the available evidence while still delivering a practically useful tool.

3.3 Retrieval-Augmented Generation for Agricultural Advisory

Direct LLM-based agricultural advice introduces significant safety concerns. In a farming context, hallucinated pesticide names, incorrect dosage suggestions, or regionally inappropriate recommendations can lead to crop damage and financial loss. Retrieval-Augmented Generation offers a safer pattern by retrieving relevant passages from verified documents before generation. This literature directly informed our system design: rather than treating the LLM as a standalone expert, we use it as a language layer over trusted source material.

3.4 Multilingual NLP and Speech Systems

A farmer-facing agricultural assistant must work across languages and input modalities. Research on multilingual embeddings, multilingual instruction-following models, and speech systems such as Whisper shows that cross-lingual semantic retrieval and multilingual transcription are feasible without building a full translation pipeline for every language pair. This was especially relevant to our use case because most agricultural source documents are in English, while users may query in Indian languages.

3.5 Gap Analysis from Existing Solutions

The review of available agritech platforms and research prototypes highlighted recurring gaps:

- Image diagnosis without conversational follow-up
- Advisory systems without image understanding
- Generic chatbot responses without grounding
- Weak support for multilingual voice interaction
- Limited attention to low-cost or constrained deployment.

These gaps shaped the project direction toward a unified multimodal assistant rather than a standalone model benchmark.

4. DATASET AND METHODOLOGY

4.1 Dataset Overview

The project uses the Top Agriculture Crop Disease dataset from Kaggle. It contains 13,324 images across 17 classes spanning 5 crops.

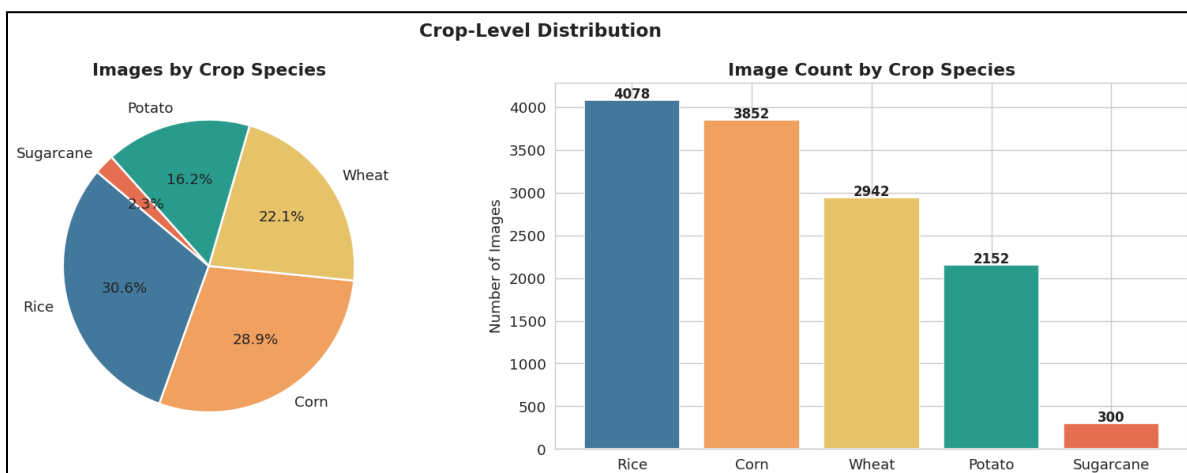
Property	Value
Dataset	Top Agriculture Crop Disease
Source	Kaggle (`kamal01/top-agriculture-crop-disease`)
Total Images	13,324
Classes	17
Crops	Corn, Potato, Rice, Wheat, Sugarcane
Task	Multi-class image classification
Image Type	RGB leaf images in JPEG / PNG formats

4.2 Crop and Class Composition

The dataset aggregates multiple sources with different capture conditions:

- Corn and Potato: PlantVillage, mostly controlled background images
- Rice: Dhan-Shomadhan and related Kaggle sources, with more natural backgrounds
- Wheat: field-style photographs with greater background variability
- Sugarcane: small, field-collected subsets with only 100 images per class

This diversity improved realism but also created major data challenges, especially inconsistency in scale, background, and class balance.

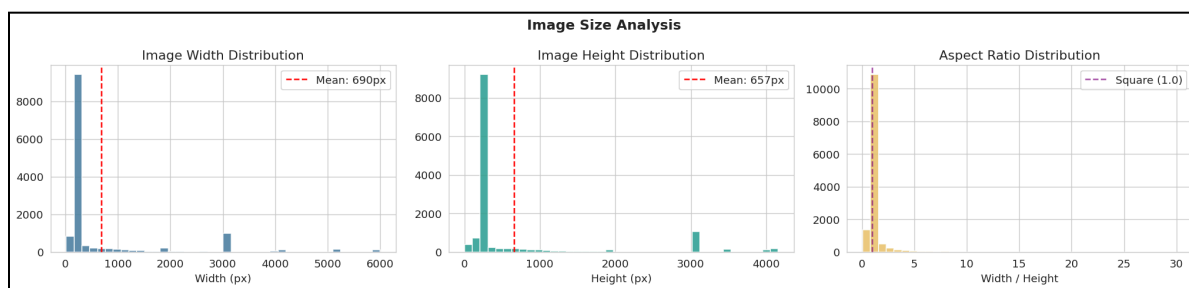


4.3 Exploratory Data Analysis

EDA was performed to understand the dataset before training. The key findings were:

- Severe class imbalance, with a 14.9x gap between the largest and smallest classes
- Extreme image size variability, from 16x1 to 6000x4160 pixels
- Consistent RGB structure across all samples
- No corrupt files and no exact duplicates
- Evidence that color shift, especially reduction in green-channel dominance, is a useful signal for disease discrimination.

These findings were not treated as descriptive only; they directly informed every preprocessing decision that followed.



4.4 Preprocessing Pipeline

The preprocessing pipeline standardized the dataset for modeling and ensured reproducibility:

Step	Technique	Purpose
Resize	224x224 using LANCZOS	Standardized input size for pretrained CNNs
RGB conversion	Forced 3-channel conversion	Prevented edge-case loading issues
Duplicate check	MD5 + perceptual hash	Reduced leakage risk
Corrupt file check	PIL verification	Prevented training failures
Normalization	ImageNet mean/std	Matched pretrained backbone expectations
Split	Stratified 80/10/10	Preserved class representation
Augmentation	Minority classes only	Reduced imbalance without contaminating evaluation

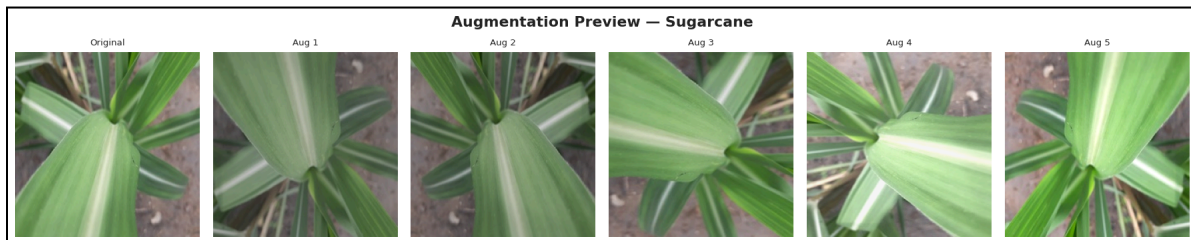
4.5 Augmentation Strategy

Augmentation was applied only to the training split and primarily targeted minority classes such as the sugarcane categories, Potato___Healthy, and Corn___Gray_Leaf_Spot. The transformations included:

- Horizontal and vertical flips

- Rotations
- Brightness and contrast shifts
- Saturation variation
- Random zoom-style crops.

This brought minority classes closer to approximately 700 training samples and materially improved fairness across classes.



4.6 Data Split and Pipeline Verification

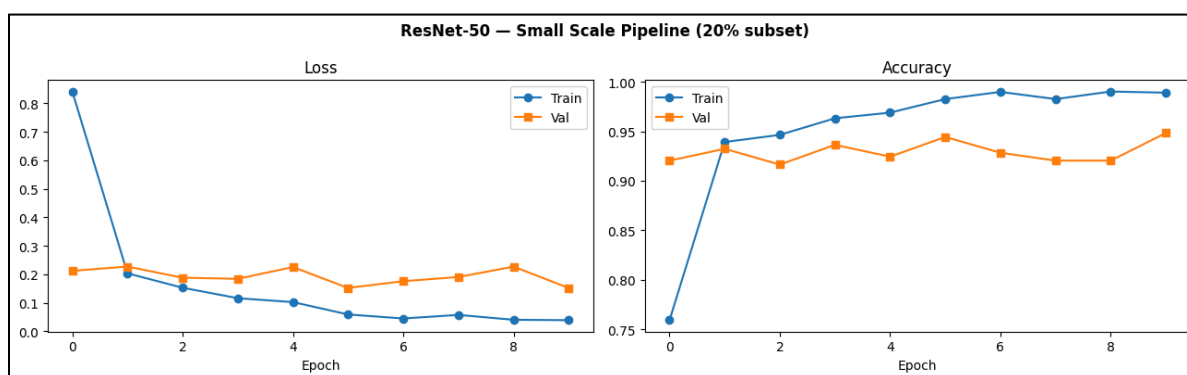
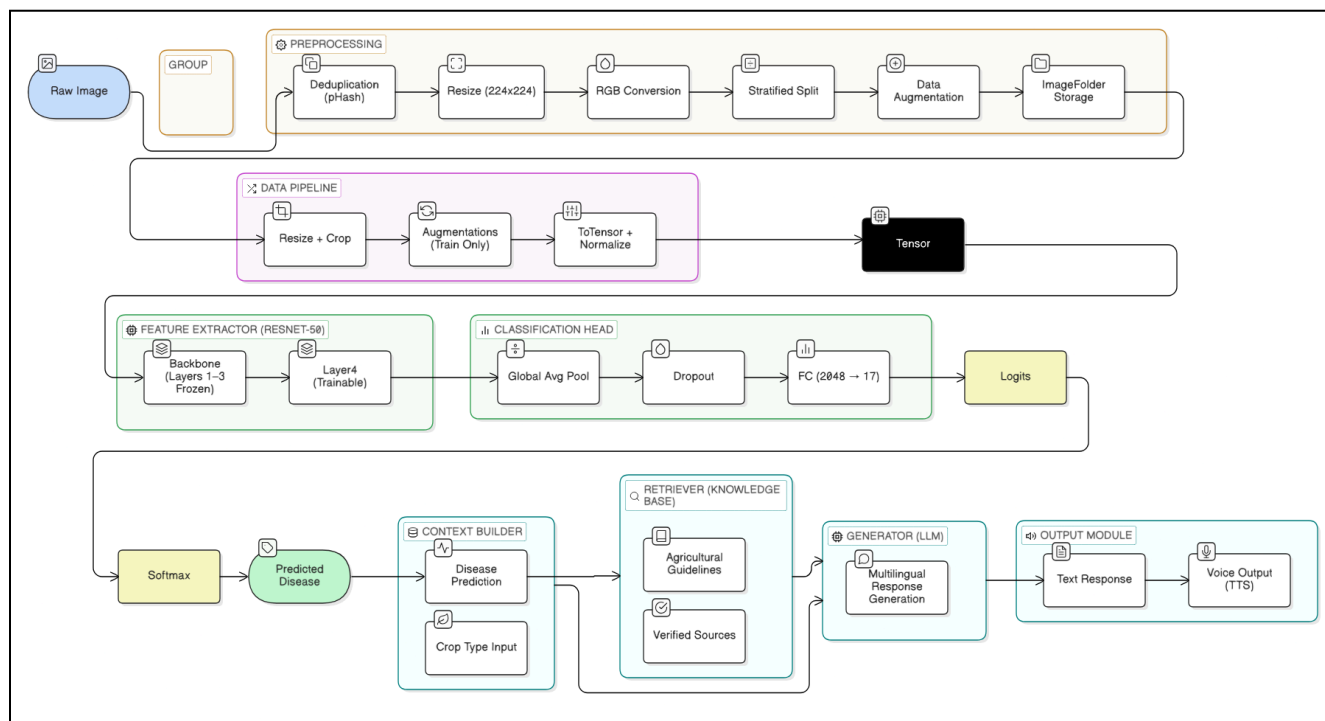
The final split was:

- Training: about 10,659 images
- Validation: about 1,332 images
- Test: about 1,333 images

The processed dataset was stored in an ImageFolder-compatible structure for direct use with PyTorch. Milestone 3 verified the end-to-end data flow from raw image through transforms, tensor conversion, forward pass, and class mapping.

4.7 Initial Architecture Exploration

Milestone 3 focused on validating the pipeline and comparing candidate model families. An early ResNet-based setup was used to validate the training flow, confirm compatibility with the processed dataset, and visualize the classification pipeline. This stage established the modeling baseline and prepared the project for systematic tuning in Milestone 4.



5. MODEL DEVELOPMENT AND HYPERPARAMETER TUNING

5.1 Model Selection Rationale

The final system required a strong balance between predictive performance and deployment efficiency. Although larger architectures such as ResNet-50 can deliver higher raw accuracy, their parameter count and inference cost are harder to justify for low-cost deployment. For this reason, the project focused on MobileNet V2 and MobileNet V3 Large, treating model family as a tunable hyperparameter rather than locking into a single architecture from the outset.

5.2 Transfer Learning Strategy

Both candidate models were initialized with ImageNet-pretrained weights. The backbone was frozen and only the classifier head was trained. This strategy was chosen because:

- ImageNet features transfer well to leaf texture and color patterns,
- the dataset, while useful, is not large enough to safely fine-tune the full network without higher overfitting risk,
- frozen-backbone training reduces compute time and supports faster experimentation.

5.3 Hyperparameter Optimization with Optuna

Optuna was used to search over:

- Architecture: MobileNet V2 or MobileNet V3 Large
- Optimizer: Adam, AdamW, SGD
- Learning rate
- Dropout
- Batch size
- Weight decay

The search used Macro F1 as the objective because overall accuracy would hide poor performance on minority classes such as sugarcane. A MedianPruner was used to stop weak runs early and reduce wasted compute.

5.4 Best Trial

The best-performing configuration was:

Parameter	Value
Architecture	MobileNet V3 Large
Optimizer	AdamW
Learning Rate	4.28e-4
Dropout	0.4704
Batch Size	16
Weight Decay	1.8e-6
Best Validation Macro F1	0.9255

5.5 Why MobileNet V3 Large Won

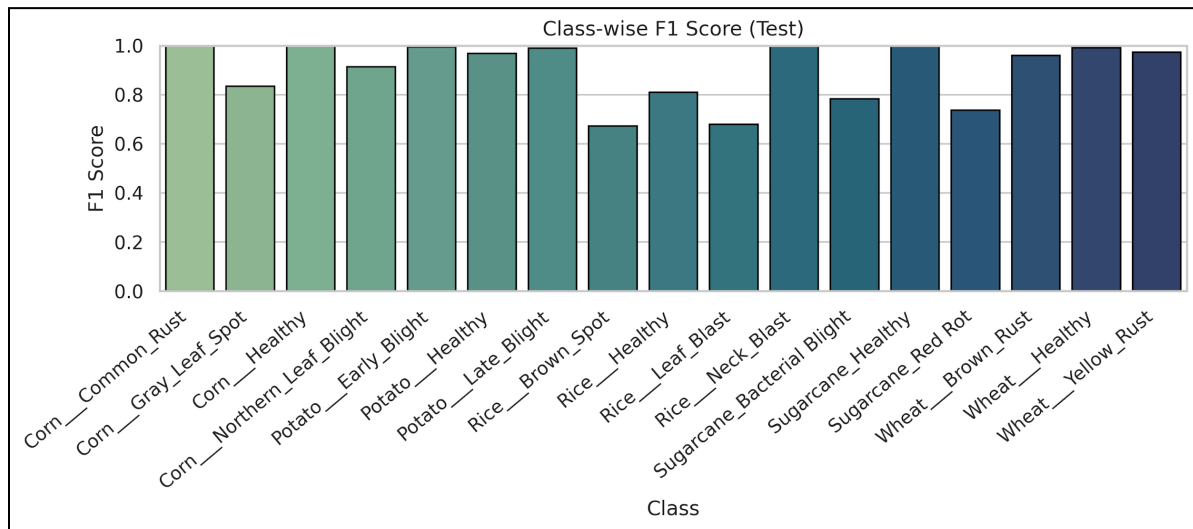
MobileNet V3 Large outperformed V2 because it offered a better trade-off between efficiency and representational power. Its use of Squeeze-and-Excitation blocks improved attention to channel-wise feature importance, which is useful in crop disease classification where texture, lesion color, and local contrast all matter.

5.6 Training Outcome

The tuned model achieved:

Metric	Value
Test Accuracy	92.09%
Test Macro F1	90.04%
Validation Macro F1	92.55%
Test Loss	0.2075

These results show that the model generalized well and retained balanced performance despite the original class imbalance.



5.7 Engineering Decisions that Improved Stability

The strongest gains came not from a single trick, but from a combination of engineering choices:

- Minority-class augmentation to reduce collapse on rare classes
- Macro F1-driven tuning rather than accuracy-driven tuning
- High dropout to reduce memorization of limited minority samples
- AdamW regularization
- Exact alignment between training and inference preprocessing.

Together, these decisions produced a model that was not only accurate but also suitable for safe downstream use in the advisory system.

6. EVALUATION AND ANALYSIS

6.1 Evaluation Setup

The final classifier was evaluated on the held-out test split only after model selection was complete. This preserved the integrity of the evaluation and prevented leakage from hyperparameter tuning.

Evaluation Item	Value
Test split size	~1,332 images
Classes	17
Primary metric	Macro F1
Secondary metrics	Accuracy, Micro F1, loss, class-wise F1, confusion matrix
Runtime	Google Colab with NVIDIA T4

6.2 Quantitative Performance

The final test performance of the best model was:

- Accuracy: 92.09%
- Macro F1: 90.04%
- Micro F1: 92.09%
- Loss: 0.2075

The relatively small gap between Macro F1 and Micro F1 indicates that the model did not achieve high performance only by dominating majority classes; it maintained reasonably balanced behavior across the label space.

6.3 Class-Wise Analysis

The model performed especially well on:

- Most corn classes
- Most wheat classes
- Potato diseases
- Sugarcane__Healthy.

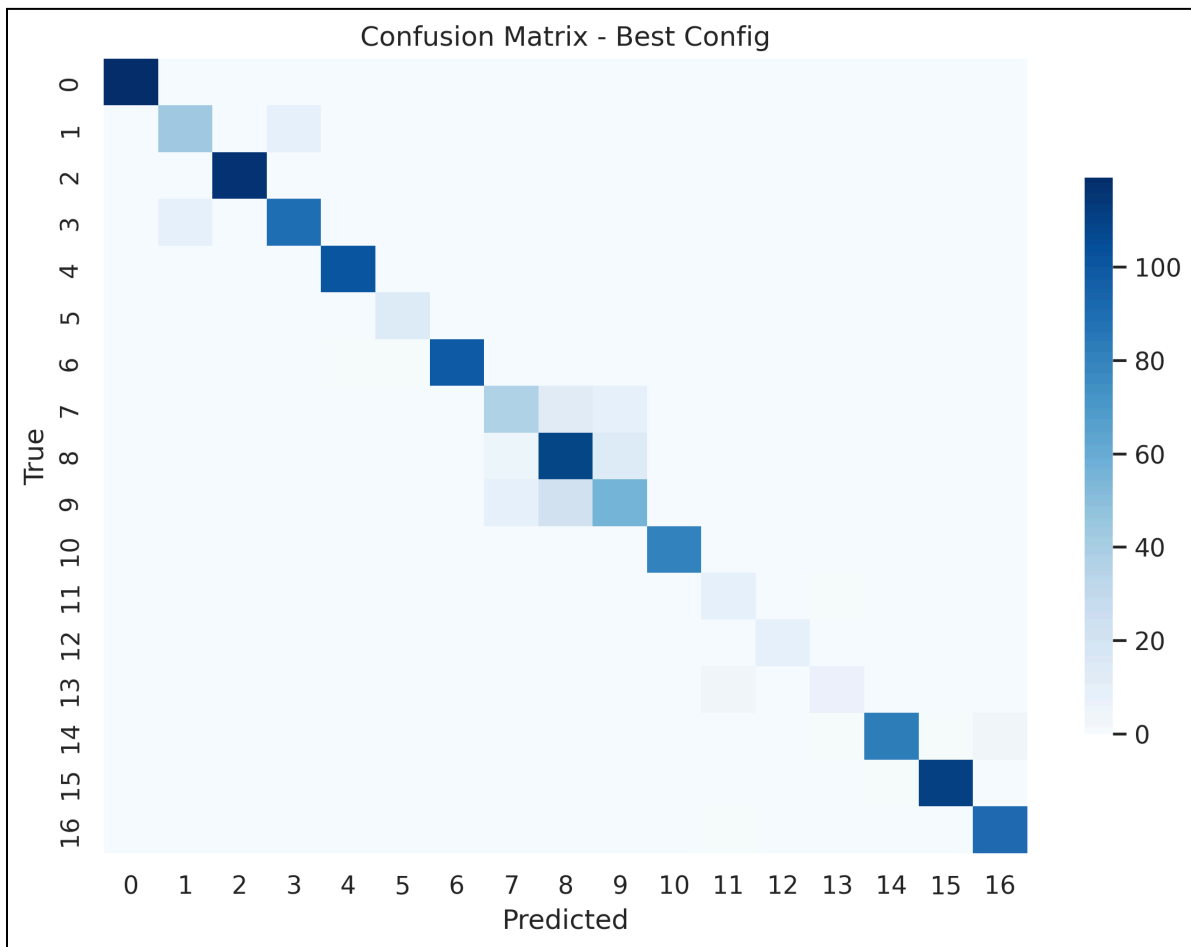
The most difficult classes were:

- Rice__Brown_Spot
- Rice__Leaf_Blast
- Rice__Neck_Blast
- Sugarcane__Red_Rot.

This pattern matches domain intuition. The rice disease categories share similar lesion morphology and color distribution, making fine-grained separation difficult. Sugarcane errors were driven more by data scarcity and limited diversity than by architectural weakness alone.

6.4 Confusion Patterns

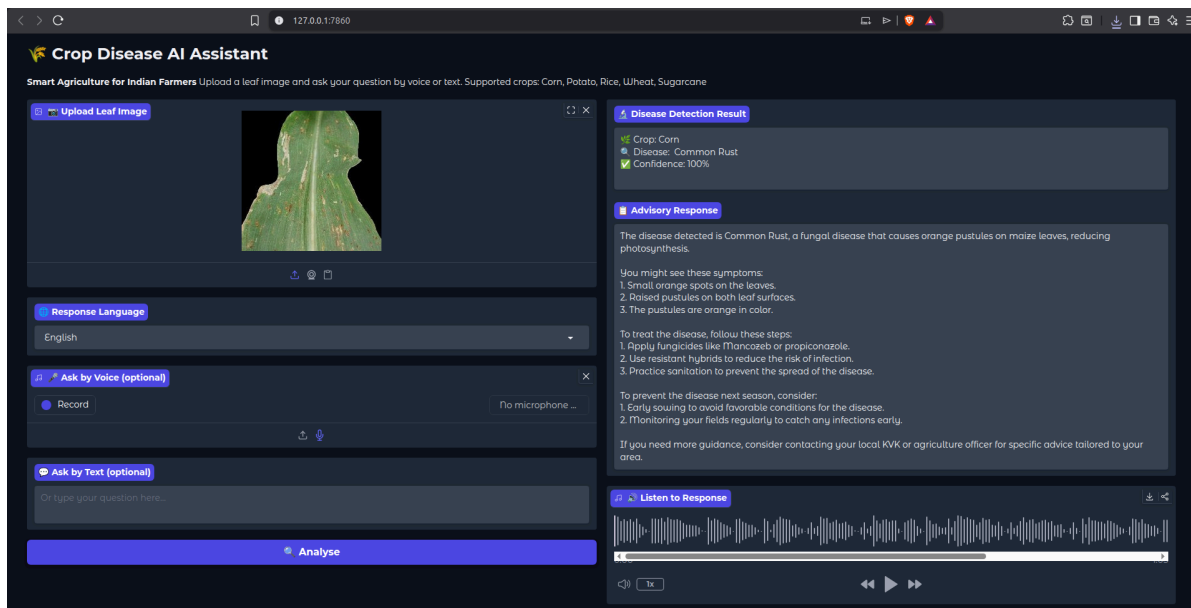
The confusion matrix shows strong diagonal dominance and very limited cross-crop confusion. In other words, the model usually identifies the crop correctly and fails mainly within closely related disease categories of the same crop rather than confusing wheat with corn or rice with potato. This is encouraging because it suggests that the vision backbone learned strong crop-level structure even when disease-level separation was harder.



6.5 Qualitative Success Cases

Milestone 5 documented strong successful predictions, including:

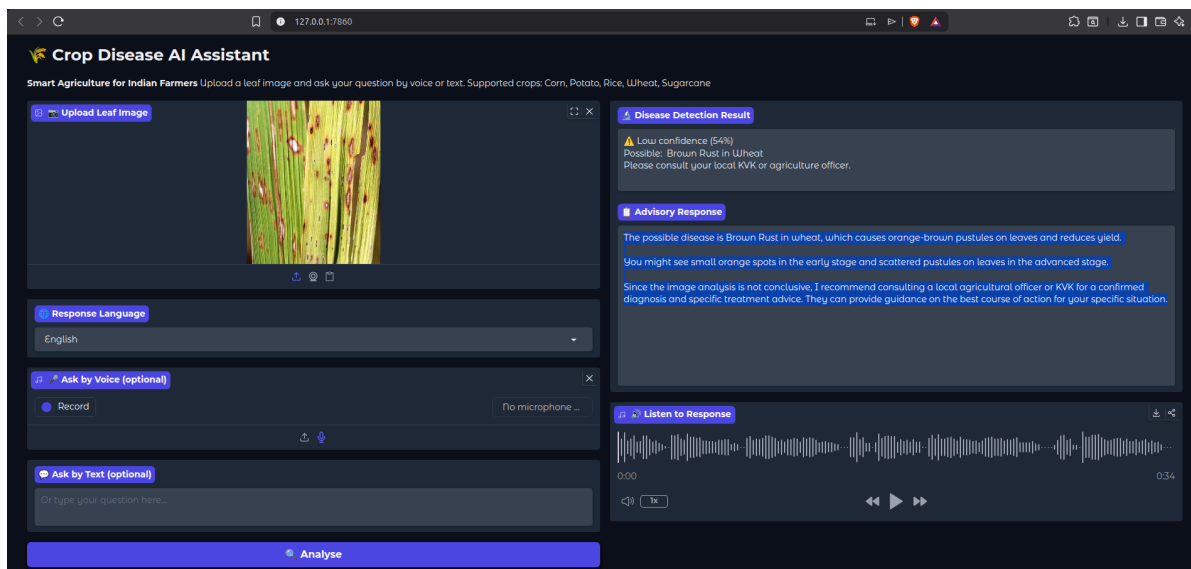
- High-confidence detection of Corn Common Rust with grounded advisory output
- Correct prediction of the minority class Sugarcane Red Rot with high confidence
- Matching text and audio responses through the full multimodal pipeline.



6.6 Failure Analysis and Safety Guardrails

A key strength of the project is that failure analysis was incorporated into the system behavior, not left only in the report. Misclassifications were most likely on visually ambiguous or noisy inputs, especially rice disease images and field photos with clutter or water droplets.

One documented failure case involved a rice leaf that the system predicted incorrectly with only 54% confidence. Because the confidence was below the 0.60 threshold, the pipeline switched to cautious mode and recommended consultation with a KVK or agricultural officer instead of giving a strong treatment recommendation. This design reduced the practical risk of unsafe advice.



6.7 Key Evaluation Insights

The evaluation phase led to three major conclusions:

1. MobileNet V3 Large is sufficiently strong for real deployment under the given constraints.
2. The main residual weakness is intra-crop fine-grained confusion, especially among rice diseases.
3. Confidence thresholding is essential because it converts model uncertainty into safer user-facing behavior.

7. DEPLOYMENT AND DOCUMENTATION

7.1 Deployed System Overview

The final application integrates image classification, RAG retrieval, LLM-based advisory generation, speech recognition, and speech synthesis in a Gradio interface deployed on Hugging Face Spaces.

Live application: <https://huggingface.co/spaces/harishsahadev/crop-disease-assistant>

7.2 End-to-End Pipeline

The deployed pipeline works as follows:

1. The user uploads a crop leaf image.
2. The MobileNet V3 Large model predicts crop, disease, and confidence.
3. If voice input is provided, Whisper transcribes the query and detects language.
4. A context builder converts the prediction into a structured retrieval query.
5. ChromaDB retrieves the most relevant chunks from verified agricultural documents.
6. Llama 3.3 70B via Groq generates an advisory constrained to the retrieved evidence.
7. gTTS converts the advisory into speech in the response language.
8. Gradio displays the diagnosis, confidence, advisory text, and playable audio.

7.3 Retrieval-Augmented Generation Stack

The RAG subsystem uses:

- Verified source documents from ICAR, TNAU, and FAO
- PyMuPDF for extraction
- Metadata-aware chunking and storage
- Intfloat/multilingual-e5-large embeddings
- ChromaDB as the local vector store.

Retrieval follows a three-stage fallback strategy:

- Crop + disease filtering first
- Crop-only filtering next
- Full semantic retrieval as fallback.

This structure improves precision and reduces cross-disease contamination in retrieved context.

7.4 Voice and Multilingual Features

The speech stack was designed for accessibility:

- Whisper-small supports multilingual ASR with automatic language detection,
- gTTS provides text-to-speech output
- The system supports English, Hindi, Bengali, Tamil, Telugu, Malayalam, and Kannada
- Voice interaction helps users who are more comfortable speaking than typing.

7.5 Deployment Constraints and Design Trade-Offs

The entire system was deployed under free-tier constraints, which required careful architectural choices:

- Lightweight CNN rather than a heavier vision backbone
- Whisper-small instead of larger ASR variants
- Local vector store instead of a paid retrieval backend
- Groq free-tier inference for the LLM
- Hugging Face Spaces for zero-cost hosting.

This made the project not only technically complete but also economically realistic as a demonstration system.

7.6 Documentation Delivered

Milestone 6 also formalized the supporting documentation required for final submission and future maintenance:

- User Guide for end users
- Developer Guide for reproducibility and system understanding
- Non-Technical Report for broad communication
- Final Technical / Project Report for submission
- Presentation materials for final review.

7.7 User-Facing Safety Measures

The deployed application includes several safeguards:

- Confidence thresholding with low-confidence warning
- KVK referral when the model is uncertain
- RAG grounding to prevent unsupported treatment advice
- Document-based fallback messaging when retrieval is weak
- Explicit limitation to supported crops and diseases.

These safeguards are especially important because agricultural decisions can carry economic and environmental consequences.

8. CONCLUSION AND FUTURE WORK

8.1 Conclusion

This project successfully delivered a multimodal AI assistant for smart agriculture that combines visual crop disease diagnosis, grounded advisory generation, multilingual interaction, and voice accessibility in a single deployed system. Across six milestones, the work progressed from problem definition and literature review to dataset curation, preprocessing, modeling, hyperparameter optimization, evaluation, deployment, and documentation.

The final system achieved 92.09% test accuracy and 90.04% Macro F1 on a 17-class crop disease task while remaining lightweight enough for free-tier deployment. More importantly, the project did not stop at prediction accuracy: it addressed trust and usability through RAG-based advisory, multilingual support, and confidence-aware safety behavior. This makes the solution much closer to a usable field-support tool than a standalone academic classifier.

8.2 Limitations

Despite the strong outcome, several limitations remain:

- The dataset is still relatively small for some classes, especially sugarcane
- Rice disease categories remain difficult to separate reliably
- The current system supports only five crops and 17 conditions
- Knowledge base is limited to the ingested document collection
- The deployed system still depends on internet access for practical use.

8.3 Future Work

The most meaningful next steps are:

1. Expand the dataset with more real field images, especially minority classes and noisy mobile captures.
2. Add more crops, diseases, and region-specific agronomic advisories.
3. Explore partial backbone unfreezing or fine-grained architectures for difficult rice classes.
4. Improve confidence calibration and threshold selection using larger real-world validation sets.
5. Add geolocation-, season-, and region-aware retrieval for more context-sensitive recommendations.
6. Investigate offline or edge-assisted inference for low-connectivity deployment.
7. Introduce source citation display in the UI so users can see which documents informed each advisory.

8.4 Final Remark

The project demonstrates that a practical agricultural AI assistant can be built responsibly when performance, grounding, accessibility, and deployment constraints are treated as part of the same system design problem. That integration, rather than any single model component, is the central achievement of the work.

Team Signatures

Name	Roll	Signature
Harish Sahadev M	21f1005856	
Sai Naman	22f3000350	
Ayushi Dixit	22f3000082	
Allanki Saketh Kumar	21f3002277	
Manas Rastogi	22f3001477	